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GTIIT

Calculus 1 - M
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Power Series

Geometric Series

Is the sum of the powers of a given number $r \in \mathbb{R}$ and only convergent for $|r| < 1$

$$\boxed{\sum_{n=0}^{\infty} r^n = \frac{1}{1-r} \; : \; |r| < 1}$$

$$\boxed{\sum_{n=k}^{\infty} r^n = \frac{r^k}{1-r} \; : \; |r| < 1}$$

The last is obtained from the first:

$$\sum_{n=k}^{\infty} r^n = r^k \cdot \sum_{n=0}^{\infty} r^n = r^k \frac{1}{1-r}$$

Power Series

1. Definition. A power series is a series of the form

$$f(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)^2 + \dots + c_n(x - x_0)^n + \dots$$

where x is a variable and the c_n are the coefficients.

- For each fixed x, the series is a numerical series, that we can test for convergence or divergence.
- The result of the series is a function

$$f(x) = \sum_{n=0}^{\infty} c_n (x - x_0)^n$$

whose domain is the set of all x for which the series converges.

- The advantage of representing a function f as a power series is that the study of f will be similar to a polynomial.

Example. Geometric Series. a) Let f be the series with $c_n = 1\ \forall n$:

$$f(x) = 1 + x + x^2 + ... + x^n$$

it is a geometric series $\sum_{n=0}^{\infty} r^n$ convergent only for $|r| < 1$:

$$f(x) = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} : \ |x| < 1$$

Let $g(x)$ be a geometric series with $r = x + 1$, centered at $x = -1$:

$$g(x) = 1 + (x+1) + (x+1)^2 + ... + (x+1)^n$$

$$= \sum_{n=0}^{\infty} (x+1)^n = \frac{1}{1-(x+1)} = -\frac{1}{x} : |x+1| < 1$$

So, the function g is defined only on $(-2, 0)$ ✓

3. EXAMPLE. Find all the values of x such that the series is convergent

$$f(x) = \sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$$

$a_n = \frac{(x-3)^n}{n}$ so Ratio test says $L < 1$ implies convergence:

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{\frac{(x-3)^{n+1}}{n+1}}{\frac{(x-3)^n}{n}} \right| = \lim_{n\to\infty} \frac{n}{(n+1)} |x-3| = |x-3| = L$$

$\Rightarrow$ it $\boxed{\text{converges for } |x-3| < 1}$ and $\boxed{\text{diverges for } |x-3| > 1}$

That is:

- The series converges on $(2,4)$
- The series diverges on $(-\infty,2) \cup (4,\infty)$
- The values for which $|x-3| = 1$ must be analyzed separately.

Power Series

Endpoints of the interval: $x = 4$

- For $x - 3 = 1 \iff x = 4$

$\Longrightarrow$ the series is an divergent positive series:

$$f(4) = \sum_{n=0}^{\infty} \frac{(x-3)^n}{n}\Big|_{x=4} = \sum_{n=0}^{\infty} \frac{1}{n} = \infty \checkmark$$

ENDPOINTS OF THE INTERVAL: $x = 2$

- For $x - 3 = -1 \iff x = 2 \implies$ the series is an alternating convergent series with

- $b_n = 1/n > 0$ decreasing,

- $\lim_{n\to\infty} \frac{1}{n} = 0$:

$$\implies f(2) = \sum_{n=0}^{\infty} \frac{(x-3)^n}{n}\Big|_{x=2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n} \text{ convergent}$$

Hence, the power series converges on $[2, 4)$ ✓

4. Theorem. For a given power series $\sum_{n=0}^{\infty} c_n (x-a)^n$ there are only three possibilities:

1. The series converges only when $x = a$
2. The series converges for all $x \in \mathbb{R}$
3. There is $R > 0$ such that the series converges if $|x-a| < R$ and diverges if $|x-a| > R$. ✓

• The number $R > 0$ is called the radius of convergence.

• In cases 1 and 2, let $R = 0$ and $R = \infty$, respectively.

• The interval of convergence consists of all x for which the series converges. It can be:
$(a-R, a+R)$, $[a-R, a+R)$, $(a-R, a+R]$ or $[a-R, a+R]$.

Radius of Convergence

- The previous theorem only speaks of the radius of convergence but it doesn't say if the series converges or not at the endpoints of the interval:

- The interval of convergence consists of all x for which the series converges. It can be:
$(a-R, a+R)$, $[a-R, a+R)$, $(a-R, a+R]$ or $[a-R, a+R]$.

5. EXAMPLE. Find all the values of x such that the series is convergent

$$f(x) = \sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}}$$

Solution: • Use the Ratio test to find R:

$$\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} = \frac{\frac{|-3|^{n+1}|x|^{n+1}}{\sqrt{n+2}}}{\frac{|-3|^n|x|^n}{\sqrt{n+1}}} = \lim_{n\to\infty} \frac{3^{n+1}|x|^{n+1}}{3^n|x|^n} \frac{\sqrt{n+1}}{\sqrt{n+2}}$$

$$= \lim_{n\to\infty} 3|x| \left(\frac{\sqrt{n+1}}{\sqrt{n+2}}\right)^{\to 1} \boxed{= 3|x| < 1 \Longrightarrow |x| < \frac{1}{3} = R}$$

Evaluate at the endpoints:

- $x = \frac{1}{3} \Longrightarrow f(\frac{1}{3}) = \sum_{n=0}^{\infty} \frac{(-3)^n \frac{1}{3^n}}{\sqrt{n+1}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$ convergent by the Alternating Series Test.

- $x = -\frac{1}{3} \Longrightarrow f(-\frac{1}{3}) = \sum_{n=0}^{\infty} \frac{(-3)^n \frac{1}{(-3)^n}}{\sqrt{n+1}} = \sum_{n=0}^{\infty} \frac{1}{\sqrt{n+1}}$ which is divergent being ot type $p = \frac{1}{2}$.

Hence, the interval of convergence is $(-\frac{1}{3}, \frac{1}{3}]$ ✓

Representation of Functions as Power Series

Represent $f(x)$ as power series on $(a-R, a+R)$ means:

- To find a series which converges to the given $f(x)$ at each $x \in (a-R, a+R)$
- Use the geometric series and algebraic manipulation
- Integrate a series
- Derivate the series
- Compositions

6. EXAMPLE. Represent $f(x) = \frac{1}{1-x}$ and $g(x) = \frac{1}{(1-x)^2}$ as power series on $(-1, 1)$

Solution: • f is the the result of a geometric series:

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} = f(x) : \ |x| < 1$$

$$(x) = \frac{1}{(1-x)^2} = \frac{d}{dx}\left[\frac{1}{1-x}\right] = \frac{d}{dx}\left[\sum_{n=0}^{\infty} x^n\right] = \sum_{n=0}^{\infty}\left[\frac{d}{dx}x^n\right]$$

$$= \sum_{n=1}^{\infty}\left[n \cdot x^{n-1}\right] = \sum_{k=0}^{\infty}\left[(k+1) \cdot x^k\right] : |x| < 1 \checkmark$$

eometric series converges for $|x| < 1$ and diverges for $|x| \geq 1$

$\boxed{\Longrightarrow R = 1 \text{ and the interval of convergence is } (-1, 1) \checkmark}$

7. EXAMPLE. Represent $h(x) = \dfrac{x^3}{(1-x)}$ as power series on $(-1, 1)$

Solution: Find a relationship with a known function

- $h(x) = \dfrac{x^3}{1-x} = x^3 \cdot \left(\dfrac{1}{1-x}\right)$

$$= x^3 \cdot \left(\sum_{n=0}^{\infty} x^n\right) = \sum_{n=0}^{\infty} x^{n+3}$$

Let $k = n + 3 : \; 3 \leq k$

$$\Longrightarrow \boxed{h(x) = \sum_{k=3}^{\infty} x^k : \quad |x| < 1 \checkmark}$$

Differentiation and Integration of Power Series.

8. Theorem. If the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ has radius of convergence $R>0$, then the function

$$f(x)=\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$$

is differentiable (and continuous) on $(a-R, a+R)$ with

1. $$f'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \dots = \sum_{n=1}^{\infty} nc_n(x-a)^{n-1}$$
2. $$\int f(x)dx = C + c_0(x-a) + c_1\frac{(x-a)^2}{2} + c_2\frac{(x-a)^3}{3} + \dots = C + \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{(n+1)}$$
3. The series converge $\forall |x-a|<R$ and diverge $\forall |x-a|>R$

Taylor Series

The previous and the following theorem are also true if $R = \infty$, for which the interval of convergence is

$$(-\infty,\ \infty)\ \checkmark$$

9. Taylor Series

Definition and Theorem. If f has a power series representation centered at a, then for $|x-a|<R$:

$$f(x)=\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \ldots$$

The coefficients are given by

$$c_n = \frac{f^{(n)}(a)}{n!}$$

where $f^{(n)}(a)$ denotes the value of the n-th derivative of f evaluated at $x=a$. So, for $|x-a|<R$,

$$f(x)=f(a)+f'(a)(x-a)+\frac{f''(a)}{2!}(x-a)^2+\ldots$$

Mc Laurin Series

10. Definition and Theorem If f has a power series representation centered at $a = 0$, then for $|x| < R$:

$$f(x) = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \ldots$$

The coefficients are given by

$$c_n = \frac{f^{(n)}(0)}{n!}$$

where $f^{(n)}(0)$ denotes the value of the n-th derivative of f evaluated at $a = 0$. The Taylor Series is called Mc Laurin series and for all $|x| < R$:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \ldots$$

11. Example Find a power series representation for $f(x) = \ln(x+1)$ on $|x| < 1$.

ution. • We notice that the derivative of $f(x) = \ln(x+1)$ is $f'(x) = \frac{1}{x+1}$ for $x+1 > 0$ and its series is a geometric series:

$$f'(x) = \frac{1}{x+1} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty} (-1)^n x^n$$

• Integrate both sides:

$$f(x) = \ln(x+1) = \int \frac{1}{x+1} dx = \int \left[1 - x + x^2 - x^3 + \ldots\right] dx$$

$$= x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \ldots + C = \sum_{n=0}^{\infty} (-1)^n \frac{1}{(n+1)} x^{n+1} + C : \ |x| < 1$$

Attention to the value of the constant of integration

This is not any antiderivative of the series, but the given function $f(x) = \ln(1+x)$, and both sides are equal, so

$$f(1) = \ln(1) = 0$$

Hence, for all $\boxed{|x| < 1}$

$$\boxed{\ln(x+1) = f(1) + \sum_{n=0}^{\infty}(-1)^n \frac{1}{(n+1)} x^{n+1} = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} x^k}$$

- For $x = 1$, the alternating series is convergent, and its value is $\ln(2)$:

$$\ln(2) = \ln(x+1)\Big|_{x=1} = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}\, x^k\Big|_{x=1} = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}$$

Hence,

$$\boxed{\ln(x+1) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}\, x^k : \quad x \in (-1, 1] \checkmark}$$

12. Example. Find a power series representation for $f(x) = \arctan(x)$ on $|x| < 1$.

Solution. • We notice that the derivative of this function is $f'(x) = \frac{1}{1+x^2}$, so we use the geometric series with $r = -x^2$, convergent for $|-x^2| < 1$:

$$f'(x) = \frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-1)^n x^{2n} : \; |x| < 1$$

• Integrate both sides:

$$\boxed{f(x) = \arctan(x)} = \int \frac{1}{1+x^2} dx = \int \left[1 - x^2 + x^4 - x^6 + \ldots\right] dx$$

$$= \arctan(0) + x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \ldots \boxed{= \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)} x^{2n+1} : \; |x| < 1 \checkmark}$$

- Let's investigate the question: under what conditions is a function equal to the sum of its Taylor series?

- If f has derivatives of all orders, when is it true that $f(x)$ is equal to its Taylor series?

- As with any convergent series, this means that $f(x)$ is the limit of the sequence of partial sums, called the nth-degree Taylor polynomial of f at a, denoted by $T_n(x)$:

$$T_n[f(x)] = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \ldots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

- Denote by $R_n(x)$ the remainder, that is, the difference between both: $R_n(x) = f(x) - T_n(x)$

Taylor Polynomial and Taylor Series

$$f(x) = \underbrace{f(a) + f'(a)(x-a) + \ldots + \frac{f^{(n)}(a)}{2!}(x-a)^n}_{T_n(x)} + \underbrace{\sum_{k=n+1}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k}_{R_n(x)}$$

= Taylor Polynomial + the remainder $R_n(x)$

13. THEOREM. Let $f(x)$ be a function that has derivatives of any order and $f(x) = T_n(x) + R_n(x)$, where T_n is the nth-degree Taylor polynomial of f at a, $T_n[f(x)] = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$.

$$\text{If } \boxed{\lim_{n\to\infty} R_n(x) = 0 \text{ for all } |x-a| < R}$$

then $f(x)$ is equal to the sum of its Taylor series on $(a-R, a+R)$, that is

$$\boxed{f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \ldots \checkmark}$$

14. TAYLOR INEQUALITY. Let $f(x)$ be a function that has derivatives of any order and $f(x) = T_n(x) + R_n(x)$, where T_n is the nth-degree Taylor polynomial of f at a,

$$T_n[f(x)] = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

If there exist $M > 0$ and d such that

$$|f^{(n+1)}(x)| \leq M \text{ for all } |x-a| < d$$

then the remainder of the Taylor series satisfies the inequality

$$\boxed{|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1} \text{ for all } |x-a| < d}$$

MACLAURIN SERIES

15. Lemma. Prove that

$$\boxed{\lim_{n\to\infty} \frac{1}{(n+1)!} | x |^n = 0 \text{ for all } x}$$

Proof of the Lemma 15.

Recall the following part of the Ratio test:
Let (a_n) be a sequence of non-zero real numbers. If the following limit exists and

$$\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} = L < 1$$

then $\sum_n a_n$ is an absolutely convergent series. In particular, $\lim_{n\to\infty} a_n = 0$.

As a consequence, we can prove the following:
For any fixed real number x, the following sequence goes to zero:

$$\boxed{\lim_{n\to\infty} \frac{x^n}{n!} = 0}$$

Proof. We consider the sequence $a_n = \frac{x^n}{n!}$.
If $x = 0$ we get $a_n = 0$ for all $n \geq 1$, that clearly has limit equal to zero.

If $x \neq 0$ we use the Ratio test for the series $\sum a_n = \sum \frac{x^n}{n!}$:

$$\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} = \lim_{n\to\infty} \frac{|x^{n+1}|}{(n+1)!} \frac{n!}{|x^n|}$$

$$= \lim_{n\to\infty} \frac{|x| \cancel{n!}}{(n+1) \cancel{n!}} = \lim_{n\to\infty} \frac{|x|}{(n+1)} = 0$$

Since $L = 0 < 1$, we conclude that the series converges, an so, the general term tends to zero:

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{x^n}{n!} = 0 \checkmark$$

16. a) Example. Prove that for all $x \in \mathbb{R}$, $f(x) = e^x$ is equal to the sum of its Maclaurin series.

Solution. • All the derivatives are the same:
$f(x) = e^x \Longrightarrow f^{(n)}(x) = e^x \; \forall n, x$.

• If $d > 0$ is any positive number, then

$$|x| < d \Longrightarrow |f^{(n+1)}(x)| = e^x < e^{\,d}$$

then, take $M = e^{\,d}$ and so obtain for the remainder

$$\boxed{|R_n(x)| \leq \frac{e^{\,d}}{(n+1)!}|x|^{n+1} \text{ for all } |\, x \,| < d}$$

• Notice that the same constant $M = e^{\,d}$ works for every value of n.

We had obtained: $\boxed{0 \le R_n \le \frac{e^{\,d}}{(n+1)!}|x|^{n+1} \text{ for all } |\, x \,| < d}$

- Moreover, by the Lemma 15:

$$\boxed{\lim_{n\to\infty} \frac{e^{\,d}}{(n+1)!}|\, x \,|^{n+1} = e^{\,d} \cdot \lim_{n\to\infty} \frac{|x|^{n+1}}{(n+1)!} = 0 \text{ for all } |\, x \,| < d}$$

- By the Squeeze Theorem:

$$\boxed{\lim_{n\to\infty} |R_n(x)| = 0 \Longrightarrow \lim_{n\to\infty} R_n(x) = 0}$$

- The Theorem implies that then $f(x) = e^x$ is equal to the sum of its Maclaurin series on $\mathbb{R}$, so $R = \infty$:

$$\boxed{e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \text{ for all } x \in \mathbb{R} \checkmark}$$

16. b) Example. Find the Taylor series for $f(x) = e^x$ centered at $x = 2$ and the interval of convergence.

Solution. • All the derivatives are the same:
$f(x) = e^x \Longrightarrow f^{(n)}(2) = e^2 \ \forall n.$

• We may continue in the same way and prove the convergence of the Taylor series at $a = 2$ as before, or:

$$e^x = e^{(x-2)+2} = e^{(x-2)}e^2$$

• Use the part a) of this example for $(x-2)$
So, as before, the Theorem implies that then $f(x) = e^x$ is equal to the sum of its Maclaurin series on $\mathbb{R}$, so $R = \infty$:

$$\boxed{e^x = \sum_{n=0}^{\infty} \frac{e^2}{n!}(x-2)^n \quad \text{for all } x \in \mathbb{R} \checkmark}$$

17. Example. Find the Maclaurin series $f(x) = \sin(x)$ and prove that it converges to f on $\mathbb{R}$.

Solution. • Arrange the computations in two columns as follows; the derivatives are periodic in a cycle of 4:

- $f(x) = \sin(x) \Longrightarrow f(0) = 0$
- $f'(x) = \cos(x) \Longrightarrow f(0) = 1$
- $f^{(2)}(x) = -\sin(x) \Longrightarrow f(0) = 0$
- $f^{(3)}(x) = -\cos(x) \Longrightarrow f(0) = -1$
- $f^{(4)}(x) = +\sin(x) \Longrightarrow f(0) = 0$

$\Longrightarrow$ the even derivatives are zero and the odd derivatives are ± 1

MACLAURIN SERIES OF $\sin(x)$

- $f^{(2n)}(0) = 0 \ \forall n$
- $f^{(2n+1)}(0) = (-1)^n \ \forall n$

- The Maclaurin series is then

$$f(x) = \sin(x) = f(0) + f'(0)x + \frac{f^{(2)}(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \ldots$$

So, the Maclaurin series of $\sin(x)$ is:

$$\boxed{\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}x^{2n+1}}$$

MACLAURIN SERIES OF $\sin(x)$.

$$|\sin(x)| \leq 1 \text{ and } |\cos(x)| \leq 1 \Longrightarrow |f^{n+1}(x)| \leq 1 \text{ for all } x \in \mathbb{R}$$

$$\boxed{\Longrightarrow 0 \leq |R_n| \leq \frac{|f^{n+1}(x)|}{(n+1)!}|x|^{n+1} \leq \frac{1}{(n+1)!}|x|^{n+1} \text{ for all } x}$$

- Moreover, by the Lemma 15:

$$\boxed{\lim_{n\to\infty} \frac{|x|^{n+1}}{(n+1)!} = 0 \text{ for all } x}$$

- By the Squeeze Theorem:

$$\boxed{\lim_{n\to\infty} |R_n(x)| = 0 \Longrightarrow \lim_{n\to\infty} R_n(x) = 0}$$

MACLAURIN SERIES OF $\sin(x)$.

- The Theorem implies that then $f(x) = \sin(x)$ is equal to the sum of its Maclaurin series on $\mathbb{R}$, so $R = \infty$:

$$\boxed{\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \quad \text{for all } x \in \mathbb{R} \checkmark}$$

18. Find the Maclaurin series $g(x) = \cos(x)$ and prove that it converges to g on $\mathbb{R}$.

Solution. g is the derivative of sin(x):

$$\cos(x) = \frac{d}{dx}\sin(x)$$

, by the Theorem, derivate the previous series to obtain:

$$\cos(x) = \frac{d}{dx}\left[\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}x^{2n+1}\right] = \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}(2n+1)x^{2n}$$

Attention that the series of the derivative starts at $n = 0$, since the derivative of its first term is $\frac{d}{dx}x = 1 \neq 0$:

MACLAURIN SERIES OF $\cos(x)$.

- The Theorem implies that then $g(x) = \cos(x)$ is equal to the sum of its Maclaurin series on $\mathbb{R}$, so $R = \infty$:

$$\boxed{\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \quad \text{for all } x \in \mathbb{R} \checkmark}$$

19. Example. Find the Maclaurin series $h(x) = x \cdot \cos(x)$ and prove that it converges to h on $\mathbb{R}$.

Solution. The theorem allows to multiply x by the series, which will be convergent to the function for all $x \in (a - R, a + R)$, hence, in this case, for all $x \in \mathbb{R}$:

$$h(x) = x \cdot \cos(x) = x \cdot \left[\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}\right]$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n+1}$$

Ex. 20. Find the Taylor Series centered at a=-2 of the following:

a) $f(x) = \dfrac{1}{(x+5)}$ b) $f(x) = \dfrac{(x^2+4x+4)}{(x+5)}$

Solution. We look for a representation of f of the form

$$f(x) = \frac{1}{(x+5)} = \sum_{n=0}^{\infty} c_n (x+2)^n$$

Since f is a rational function, we can use a geometric series:

$$\frac{1}{(x+5)} = \frac{1}{3+(x+2)} = \frac{1}{3}\frac{1}{\left[1+\left(\frac{(x+2)}{3}\right)\right]} = \frac{1}{3}\frac{1}{\left[1-\underbrace{\left(-\frac{(x+2)}{3}\right)}_{=r}\right]} = \frac{1}{3}\frac{1}{(1-r)} = \frac{1}{3}\sum_{n=0}^{\infty} r^n$$

$$\boxed{= \frac{1}{3}\sum_{n=0}^{\infty}\left(-\frac{(x+2)}{3}\right)^n = \sum_{n=0}^{\infty}\frac{(-1)^n}{3^{n+1}}\left((x+2)\right)^n}$$

b) $f(x) = \dfrac{(x^2+4x+4)}{(x+5)} = (x^2+4x+4)\dfrac{1}{(x+5)} = (x+2)^2\dfrac{1}{(x+5)}$

$$= (x+2)^2 \sum_{n=0}^{\infty} \frac{(-1)^n}{3^{n+1}} \left((x+2)\right)^n$$

$$\boxed{= \sum_{n=0}^{\infty} \frac{(-1)^n}{3^{n+1}} (x+2)^{n+2}}$$

Change of Variables: $k = n+2$ then k starts at $k = 2$ and $(-1)^n = (-1)^{k-2} = \dfrac{(-1)^k}{(-1)^2} = (-1)^k$, then

$$\boxed{= (x+2)^2 \frac{1}{(x+5)} = \sum_{k=2}^{\infty} \frac{(-1)^k}{3^{k-1}} (x+2)^k}$$